Ahmed Reda Elshazly

Mechanical Power Engineer

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Education

Ain Shams University | Cairo, Egypt

Undergraduate, B.Sc. in Mechanical Power Engineering | CGPA: 2.7

El-Mokattam Language and International School (MILS) | Cairo, Egypt

Egyptian Secondary Education | Grade: 86.1%

Ranked 1st at school; Top 20 in the district.

Professional & Team Experience

Aero ASU | Cairo, Egypt | Link

Head of Aerodynamics Team | Feb 2025 – Present

- Spearheaded the foundation of the aerodynamics subsystem from scratch, establishing standardized engineering workflows.
- Conducted technical onboarding and elementary training sessions in aerodynamics and aircraft design fundamentals for team members.
- Led the aerodynamic design and optimization of the team's aircraft for the International Aviation Engineering Conference (ICMTC) competition.

EgyptAir Academy | Cairo Airport, Egypt | Link

Airframe and Powerplant Intern | Jul 2024 – Aug 2024

- Studied aircraft systems, subsystems, and aerodynamics principles, including HVAC, powerplants, fuel systems, airframes, landing gear, and auxiliary power units (APUs).
- Observed and learned overhaul procedures for various Boeing and Airbus commercial aircraft across Hangar 7000 and 8000.

Dynamics Distribution (Fiat & Alfa Romeo) | Al-Obour, Egypt | Link

Automotive Maintenance Intern | Sep 2024 – Sep 2024

- Participated hands-on in comprehensive vehicle diagnostics and servicing.
- Assisted with engine oil/filter changes, steering corrections, fuel pump replacements, brake drum changes, turbocharger replacements, and full engine overhauls.

Mathematics Tutor | Elmokattam, Egypt |

Independent Tutor & Content Creator | Aug 2021 – Present

- Instructed Egyptian Secondary Education and SAT students to improve test scores and simplify complex mathematical concepts.
- Co-authored a specialized mathematics textbook tailored for secondary education students.

Technical Projects

Aircraft Design for ICMTC Competition | Feb 2025 – Present | Link

- Designed a competitive aircraft using Flow5, SolidWorks, and Ansys Fluent.
- Extracted and analyzed performance curves across takeoff, cruise, and landing configurations (Power/Thrust Required, Stability Derivatives, Static Margin, COG placement, and stall behavior).

Air Powered Car | Sep 2025 – Feb 2026 | Link

- Engineered the chassis, powertrain, and propulsion system for a compressed-air vehicle.
- Utilized SolidWorks Weldments and Design Accelerator for structural design; applied GD&T and strict tolerancing for production-ready assembly drawings.
- Validated structural integrity under load using finite element analysis (FEA) simulations in Autodesk Inventor.

Pneumatic Solar Tracker | Feb 2026 – Mar 2026 | Link

- Designed the mechanical architecture of a 3-DOF solar tracker utilizing spherical and universal joints.
- Modeled full CAD assemblies and extracted manufacturing drawings using SolidWorks.
- Performed non-linear stress analysis on custom 3D-printed spherical joints to prevent structural failure.

Rocket Engine Design | Sep 2024 – Feb 2025 | Link

- Designed a first-stage supersonic rocket nozzle based on fluid mechanics principles.
- Performed stoichiometric combustion calculations and applied isentropic flow equations to establish nozzle geometry.
- Validated flow characteristics and accounted for losses using Ansys Fluent axisymmetric CFD simulations.

Cooling System for an Aluminum Block | Mar 2025 – Apr 2025 | Link

- Evaluated three thermal configurations: Free Convection, Forced Convection, and Forced Convection with Fins.
- Calculated transient free convection behavior analytically using the lumped capacitance model and validated results via Ansys Transient Thermal.

Mechanical Component Design (Hand Press, Oldham Coupling, Crane Hook) | Sep 2024 – Feb 2025 | Link

- Modeled a crane hook and Oldham coupling in SolidWorks, and a hand press in Autodesk Inventor.
- Implemented parametric modeling techniques and strict tolerance stack-ups to allow for rapid product variance.

Technical Skills

- **Aerodynamics:** Lift & Drag (C_L, C_D, C_m), Stall Behavior, Flow Separation, Downwash, Oswald Efficiency.
- **Mechanical Design & CAD:** SolidWorks, Autodesk Inventor, Technical/Working Drawings, GD&T, Tolerancing, Stress Concentration, Fatigue, Buckling.
- **CFD & Thermal Analysis:** Ansys Fluent, CFX, Turbulence Modeling (y+ verification), Mesh Independence Studies, Heat Transfer, Axisymmetric Modeling, Transient Thermal Analysis.
- **Microsoft Office:** Word, Excel "Equation, Graphs, Data" & Powerpoint
- **EES:** Engineering Equation Solver, Thermodynamic Analysis
- **Languages:** Arabic (Native), English (Fluent).

Certifications

- **CSWPA-WD:** Certified SolidWorks Professional Advanced - Weldments (Dec 2025) Certificate
- **CSWPA-SUR:** Certified SolidWorks Professional Advanced - Surfacing (Dec 2025) Certificate
- **CSWPA-SM:** Certified SolidWorks Professional Advanced - Sheet Metal (Dec 2025) Certificate
- **CSWPA-DT:** Certified SolidWorks Professional Advanced - Drawing Tools (Dec 2025) Certificate
- **CSWP:** Certified SolidWorks Professional (Nov 2025) Certificate
- **CSWA:** Certified SolidWorks Associate (Nov 2025) Certificate